Pharmaceutical Calculations Booklet

Midterm · Lectures 1 – 7

Contents

		Lecture Problems	Textbook Problems	Page
1	Introduction, Units & Equivalences	21	94	**3**
2	Drug Concentrations	15	33	**25**
3	Dilutions	11	35	**35**
4	Moles	10	40	**43**
5	Doses for Adults	9	45	**54**
6	Doses for Children	12	15	**63**
7	Creatinine Clearance	9		**68**

Each lecture is split into topics. Every topic has **Lecture Problems**, then **Textbook Problems**.

1 Introduction, Units & Equivalences

In this lecture

1. Proportional Sets & Ratios 4 lecture · 7 textbook
2. Fractions 1 lecture · 16 textbook
3. Decimals 1 lecture · 9 textbook
4. Percentages 1 lecture · 12 textbook
5. Finding a Missing Value & Practical Problems 24 textbook
6. Exponentials (Powers of 10) 1 lecture
7. Ratio Strength, Percentage Strength & ppm 3 lecture
8. Variation (Inverse Proportion) 1 lecture
9. Units, Prefixes & Equivalences 4 lecture · 15 textbook
10. Conversions Between Systems & Temperature 1 lecture · 11 textbook
11. Dosage Calculations 4 lecture

1.1 Proportional Sets & Ratios

Lecture Problems

Q1 A drug has a concentration of 100 mg/5 ml. Complete the table:

Weight of drug (mg)	Volume (ml)
20	?
50	?
75	?
150	?

a. Volume that contains 20 mg = ?

b. Volume that contains 50 mg = ?

c. Volume that contains 75 mg = ?

d. Volume that contains 150 mg = ?

Q2 A car travels at 100 km/hour. Complete the table:

Distance (km)	Time (hour)
50	?
150	?
200	?

a. Time for 50 km = ?

b. Time for 150 km = ?

c. Time for 200 km = ?

Q3 The upper set is proportional to the lower set:

9	21	30	36
3	7	10	12

What is the ratio of the upper set to the lower set?

Q4 The upper set is proportional to the lower set:

2	5	7
10	25	35

What is the ratio of the upper set to the lower set?

Textbook Problems

Q5 In each of the following, decide whether or not the pair of sets is proportional.

2.1

6	10	14
3	5	7

2.2

5	7	9
40	56	72

2.3

2	4	8
1	2	3

2.4

5	7	9
15	17	19

2.5

16	17	21
80	85	105

Q6 Decide whether or not the pair of sets is proportional.

6	8	10…
3	5	7…

Proportional or not?

Q7 Decide whether or not the pair of sets is proportional.

5	7	9…
20	28	36…

Proportional or not?

Q8 Decide whether or not the pair of sets is proportional.

2	3	4…
5	6	7…

Proportional or not?

Q9 Decide whether or not the pair of sets is proportional.

2	4	6…
5	10	15…

Proportional or not?

Q10 Decide whether or not the pair of sets is proportional.

25	35…
5	7…

Proportional or not?

Q11 Decide whether or not the pair of sets is proportional.

2	4	6…
3	6	9…

Proportional or not?

1.2 Fractions

Lecture Problems

Q12 x/3 = 16/24

Find what is x.

Textbook Problems

Q13 Find the numerator of the fraction that is equal to 1/7 but has denominator 21.

Numerator = ?

Q14 Find the numerator of the fraction that is equal to 16/24 but has denominator 3. (Same as Q12.)

Numerator = ?

Q15 Find the denominator of the fraction that is equal to 80/180, but has numerator 16.

Denominator = ?

Q16 Find the fraction that is equal to 7/8 but has denominator 72.

Answer = ?

Q17 Find the fraction that is equal to 4/7 but has numerator 28.

Answer = ?

Q18 Find the fraction that is equal to 2/9 but has denominator 117.

Answer = ?

Q19 Express 35/45 in its lowest terms.

Lowest terms = ?

Q20 Express 60/405 in its lowest terms.

Lowest terms = ?

Q21 Express 12/20 in its lowest terms.

Answer = ?

Q22 Express 3/15 in its lowest terms.

Answer = ?

Q23 Express 42/54 in its lowest terms.

Answer = ?

Q24 Express 78/130 in its lowest terms.

Answer = ?

Q25 Express 120/42 in its lowest terms.

Answer = ?

Q26 Multiply 3/13 by 5.

Answer = ?

Q27 Multiply 3/17 by 4.

Answer = ?

Q28 Multiply 5/9 by 7.

Answer = ?

1.3 Decimals

Lecture Problems

Q29 The number 0.23 is a rational number: the 2 is in the tenths column and the 3 is in the hundredths column, so 0.23 = 23/100.

a. Convert 0.5 to a fraction.

b. Convert 0.44 to a fraction.

Textbook Problems

Q30 Convert 0.3 to a fraction.

Fraction = ?

Q31 Convert 0.35 to a fraction.

Fraction = ?

Q32 Convert 0.8 to a fraction in its lowest terms.

Answer = ?

Q33 Convert 0.87 to a fraction in its lowest terms.

Answer = ?

Q34 Convert 0.875 to a fraction in its lowest terms.

Answer = ?

Q35 Convert the fraction 3/5 to a decimal.

Decimal = ?

Q36 Convert the fraction 7/8 to a decimal.

Decimal = ?

Q37 Convert the fraction 5/8 to a decimal.

Answer = ?

Q38 Convert the fraction 7/80 to a decimal.

Answer = ?

1.4 Percentages

Lecture Problems

Q39 8 percent = 8 per 100 = 8/100

a. Convert 3/8 to percent.

b. Convert 0.125 to percent.

c. Convert 25% to fraction.

d. Convert 0.6% to fraction.

Textbook Problems

Q40 Express 7/10 as a percentage.

Answer = ?

Q41 Express 3/8 as a percentage.

Answer = ?

Q42 Express 7/8 as a percentage.

Answer = ?

Q43 Express 6/25 as a percentage.

Answer = ?

Q44 Express 0.76 as a percentage.

Answer = ?

Q45 Express 0.769 as a percentage.

Answer = ?

Q46 Express 0.82 as a percentage.

Answer = ?

Q47 Express 0.345 as a percentage.

Answer = ?

Q48 Express 20% as a fraction in its lowest terms.

Answer = ?

Q49 Express 0.2% as a fraction in its lowest terms.

Answer = ?

Q50 Express 30% as a fraction in its lowest terms.

Answer = ?

Q51 Express 0.3% as a fraction in its lowest terms.

Answer = ?

1.5 Finding a Missing Value & Practical Problems

Textbook Problems

Q52 For the proportional sets:

7	y	9	10…
21	24	27	30…

Find the missing value y.

Q53 For the proportional sets:

x	14
25	50

Find the missing number x.

Q54 For the proportional sets:

3	x
5	8

Find the missing value x.

Q55 For the pair of proportional sets:

z	9
6	15

Find the missing value z.

Q56 For the proportional sets:

37	52
7	z

Find the missing number z.

Q57 Find the value of p in the two proportional sets:

36	9
22	p

p = ?

Q58 Find the missing value x from the pair of proportional sets:

8	x	20
2	4	5

Missing value = ?

Q59 Find the missing number from the pair of proportional sets:

25	30
y	6

Missing value = ?

Q60 Find the missing number from the pair of proportional sets:

21	z
7	30

Missing value = ?

Q61 Find the missing number from the pair of proportional sets:

81	18
x	2

Missing value = ?

Q62 Find the missing number from the pair of proportional sets:

y	21
8	24

Missing value = ?

Q63 Find the missing number from the pair of proportional sets:

4	10
10	z

Missing value = ?

Q64 Find the missing number from the pair of proportional sets:

x	10
7	35

Missing value = ?

Q65 Find the missing number from the pair of proportional sets:

32	y
19	41

Give your answer correct to two significant figures.

Missing value = ?

Q66 Find the missing number from the pair of proportional sets:

225	420
192	z

Give your answer correct to three significant figures.

Missing value = ?

Q67 Find the missing number from the pair of proportional sets:

x	11.3
22.5	15.7

Give your answer correct to three significant figures.

Missing value = ?

Q68 Consider three sets each proportional to each other:

x	20
y	50
30	150

Find x and y.

Q69 Find the values of a and b in the following proportional sets:

a	25	110
12	b	22

a = ?, b = ?

Q70 Find the values of x and y in the three proportional sets:

24	6
x	15
64	y

x = ?, y = ?

Q71 Find the values of y and z in the two proportional sets:

y	18	21
9	z	63

y = ?, z = ?

Q72 A suspension contains 120 mg of paracetamol in each 5 mL. Find the volume of suspension that contains 300 mg of paracetamol.

Volume = ?

Q73 The recommended dose of phenytoin for a patient is 840 mg. An injection contains 50 mg/mL. What volume of injection is needed for the recommended dose?

Volume = ?

Q74 1 L of aqueous solution contains 250 mL of syrup. How much syrup will be contained in 25 L of the solution?

Volume of syrup = ?

Q75 25 g of ointment contains 5 g of calamine. How much calamine will be contained in 60 g of ointment?

Mass of calamine = ?

1.6 Exponentials (Powers of 10)

Lecture Problems

Q76 Physical and chemical measurements deal with either very small or large numbers, so for convenience we use exponentials. Positive power means multiply; negative power means divide.

a. Write 3,000,000 using a power of 10.

---

b. Write 0.000003 using a power of 10.

1.7 Ratio Strength, Percentage Strength & ppm

Lecture Problems

Q77 A percentage is the number of parts per one hundred and is probably the commonest expression of concentration used in pharmacy.

Express 1:2500 as a percentage strength.

Q78 Express 0.1% as a ratio strength.

Ratio strength = ?

Q79 Express 1 p.p.m. as a percentage strength.

Percentage strength = ?

1.8 Variation (Inverse Proportion)

Lecture Problems

Q80 Sometimes pharmaceutical calculations deal with inverse relationships — twice the cause, half the effect — as when you decrease the strength of a solution by increasing the amount of diluent.

If 10 litres of a 5% solution are diluted to 40 litres, what is the percentage strength of the dilution?

1.9 Units, Prefixes & Equivalences

Lecture Problems

Q81 Prefixes table — complete the **Factor** column:

Prefix	Symbol	Division	Factor
mega	M	× 1000,000	?
kilo	k	× 1000	?
deci	d	÷ 10	?
centi	c	÷ 100	?
milli	m	÷ 1000	?
micro	mc	÷ 1000,000	?
nano	n	÷ 1000,000,000	?

a. mega (M)

b. kilo (k)

c. deci (d)

d. centi (c)

e. milli (m)

f. micro (mc)

g. nano (n)

Q82 Complete the equivalences:

a. 1 L = ? ml

b. 1 kg = ? g = ? mg = ? mcg = ? ng

Q83 Examples — fill in the blanks:

a. 0.5 kg = ? g = ? mg

b. 0.25 g = ? mg (and in kg?)

c. 0.2 mg = ? mcg

d. 0.4 mg = ? kg

e. 0.5 L = ? ml

f. 0.25 mol = ? mmol

Q84 For drug calculation it is best to work in whole numbers (i.e. 125 mcg not 0.125 mg). To convert from a larger unit to a smaller unit: × 1000. From a smaller unit to a larger unit: ÷ 1000.

Write 0.125 mg as a whole number in a smaller unit.

Textbook Problems

Q85 Change 200 mg to grams.

Answer = ?

Q86 Convert 5473 mL to litres.

Answer = ?

Q87 Convert 1.26 mg to micrograms.

Answer = ?

Q88 Convert 135 465 micrograms to grams.

Answer = ?

Q89 Convert 0.003 745 kL to mL.

Answer = ?

Q90 Change 0.035 g to milligrams.

Answer = ?

Q91 Change 1384 mg to grams.

Answer = ?

Q92 Change 437 mL to litres.

Answer = ?

Q93 Change 12.47 L to millilitres.

Answer = ?

Q94 Change 0.87 mL to microlitres.

Answer = ?

Q95 Change 15 750 microlitres to millilitres.

Answer = ?

Q96 Change 0.0025 kg to grams.

Answer = ?

Q97 Change 12 g to kilograms.

Answer = ?

Q98 Change 0.72 megalitres to kilolitres.

Answer = ?

Q99 Change 145 kg to megagrams.

Answer = ?

1.10 Conversions Between Systems & Temperature

Lecture Problems

Q100 Height: convert inch to cm. Weight: convert pound to kg. (Use the standard factors 1 inch = 2.54 cm and 1 pound = 0.4536 kg.)

a. Convert 10 inches to cm.

---

b. Convert 150 pounds to kg.

Textbook Problems

Q101 Convert 3.4 pints to litres. (1 L = 1.76 pints)

Litres = ?

Q102 Use the conversion factor 1 L = 1.76 pints to express:

(a) 6.43 pints in litres

(b) 2.42 L in pints

Q103 Convert 106.4 kg to stones and pounds. (1 pound = 0.4536 kg; 14 pounds = 1 stone)

Stones and pounds = ?

Q104 Convert 10 stone 4 pounds to kilograms. (1 pound = 0.4536 kg)

Kilograms = ?

Q105 Use the conversion factor 1 pound = 0.4536 kg to express:

(a) 9 stones 4 pounds in kilograms (1 stone = 14 pounds)

(b) 49 kg in stones and pounds

Q106 Convert 5 feet 10 inches into metres. (1 inch = 25.4 mm = 0.0254 m)

Metres = ?

Q107 What would be the BMI for a patient who weighs 10 stone and 4 pounds and is 5 feet 10 inches tall? (BMI = weight (kg) / height² (m))

BMI = ?

Q108 Find the number of degrees Celsius corresponding to 95 degrees Fahrenheit. (50 °C = 122 °F; water freezes at 0 °C = 32 °F)

°C = ?

Q109 Change 25 degrees Celsius to degrees Fahrenheit. (0–100 °C corresponds to 32–212 °F, a range of 100 °C and 180 °F)

°F = ?

Q110 Change 68 degrees Fahrenheit to degrees Celsius.

°C = ?

Q111 The boiling point of water is 100 degrees Celsius or 212 degrees Fahrenheit. The freezing point of water is 0 degrees Celsius or 32 degrees Fahrenheit. Express:

(a) 15 degrees Celsius in degrees Fahrenheit (to the nearest degree)

(b) 88 degrees Fahrenheit in degrees Celsius (to the nearest degree)

1.11 Dosage Calculations

Lecture Problems

Q112 The dose required is 3 mg/kg and the patient weighs 68 kg. This means that for every kg of patient's weight, you will need 3 mg of drug.

Total dose required = ?

Q113 In some cases, the dose is given as a total daily dose — it has to be given in divided doses (3 or 4 times a day).

Ibuprofen 1.2 g daily given 3 times daily. What is each single dose?

Q114 The dose for Cefuroxime iv is 2.25 g in divided doses every 8 hours.

What is each dose?

Q115 You need to prepare an infusion of digoxin containing 0.75 mg. Digoxin is available as 500 mcg/2 ml.

How many ml do you need?

---

2 Drug Concentrations

In this lecture

1. Fractions, Decimals & Percentages 2 lecture
2. Calculations with % 3 lecture
3. Ratio Strength 1 lecture · 7 textbook
4. Percentage Concentration (w/v, w/w, v/v) 3 lecture · 16 textbook
5. mg/ml Concentrations & Amount Strength 2 lecture · 2 textbook
6. Parts per Million (ppm) 2 textbook
7. Converting Between Expressions of Concentration 6 textbook
8. Heparin & Insulin (Units) 1 lecture
9. Practice Calculations (Lecture) 3 lecture

2.1 Fractions, Decimals & Percentages

Lecture Problems

Q1 Fractions & percentages: to convert a fraction to a percentage, multiply by 100; to convert a percentage to a fraction, divide by 100.

a. Convert 2/8 to a percentage.

b. Convert 25% to a fraction.

Q2 Decimals & percentages: to convert a decimal to a percentage, multiply by 100; to convert a percentage to a decimal, divide by 100.

a. Convert 0.5 to a percentage.

b. Convert 50% to a decimal.

2.2 Calculations with %

Lecture Problems

Q3 How much is 35% in 500 ml?

Amount = ?

Q4 What is the percentage of 480 of 8000?

Percentage = ?

Q5 How many ml is 40% of 1.3 litre?

Volume (ml) = ?

2.3 Ratio Strength

Lecture Problems

Q6 Ratio strength is expressed as a ratio in the form of 1 in x.

5 g of product contains 250 mg of sulphur in yellow soft paraffin. Express this in ratio.

Textbook Problems

Q7 2 L of an aqueous solution contains 50 mL of ethanol. Express this as a ratio strength.

Ratio strength = ?

Q8 5 g of product contains 250 mg of sulfur in yellow soft paraffin. Express this as a ratio strength. (Same as Q6.)

Ratio strength = ?

Q9 What is the amount of potassium permanganate in 300 mL of a 1 in 25 solution and what is the percentage strength of the solution?

Amount = ? and % = ?

Q10 How many milligrams of y are needed to make 200 mL of a 1 in 500 solution?

Answer = ?

Q11 How many millilitres of y are needed to produce 400 mL of a 1 in 200 solution?

Answer = ?

Q12 How many milligrams of y are needed to produce 25 g of a 1 in 5 ointment?

Answer = ?

Q13 How many grams of y are there in 250 mL of a 1 in 80 solution?

Answer = ?

2.4 Percentage Concentration (w/v, w/w, v/v)

Lecture Problems

Q14 The most common way of describing drug concentration is percentage concentration.

a. What does 3% w/v mean?

---

b. What does 3% w/w mean?

---

c. What does 3% v/v mean?

Q15 How much dextrose is there in a 500 ml infusion of dextrose 4% w/v?

Dextrose = ?

Q16 You have salbutamol nebuliser solution 0.3%.

What volume is required for a 4 mg dose?

Textbook Problems

Q17 How many milligrams of aluminium acetate are required to prepare 500 mL of a 0.03% w/v solution?

Amount (mg) = ?

Q18 How many milligrams of x are needed to make 5 mL of an 8% solution?

Answer = ?

Q19 How many millilitres of x are needed to make 600 mL of a 3% solution?

Answer = ?

Q20 How many milligrams of x are needed to make 200 mL of a 3.4% solution?

Answer = ?

Q21 How many milligrams of x are there in 100 mL of a 0.01% solution?

Answer = ?

Q22 How much calamine is required to produce 250 g of a 3% ointment?

Answer = ?

Q23 Potassium permanganate 0.1 g; Purified water to 100 mL.

How many millilitres of a 2.5% potassium permanganate solution could be used in place of the 0.1 g in the above formula?

Q24 What volume of 17% w/v solution contains 1.5 g of ingredient?

Answer = ?

Q25 What volume of 20% w/v solution contains 5 g of ingredient?

Answer = ?

Q26 What volume of 15% w/v solution contains 7 g of ingredient?

Answer = ?

Q27 What volume of 34% w/v solution contains 150 g of ingredient?

Answer = ?

Q28 A cream contains 12 g of drug X made up to 100 g with cream base. What is the percentage concentration?

Percentage = ?

Q29 Express 900 mg of sodium chloride made up to 100 mL with water as a percentage.

Percentage = ?

Q30 If the following amounts of drug are made up to 5 g with lactose what is the percentage concentration of the resulting mix?

(a) 100 mg

(b) 150 mg

(c) 200 mg

(d) 300 mg

(e) 500 mg

Q31 Express 1 in 500 w/v solution of potassium permanganate as a percentage.

Percentage = ?

Q32 A morphine sulfate injection contains 10 mg/mL. What is the percentage concentration?

Percentage = ?

2.5 mg/ml Concentrations & Amount Strength

Lecture Problems

Q33 Sometimes it may be useful to convert percentage concentrations to mg/ml.

Express sodium chloride 0.9% in mg/ml.

Q34 You can find the strength in mg/ml by multiplying the percentage by 10; to convert mg/ml to percentage, divide by 10.

a. Convert 0.9% to mg/ml.

b. Convert 9 mg/ml to a percentage.

Textbook Problems

Q35 A preparation contains 900 mg of sodium chloride dissolved in water to produce 100 mL of solution. Express the concentration of the solution as an amount strength (mg/mL and g/mL).

Amount strength = ?

Q36 A solution contains a solid dissolved in a liquid. The ratio strength is 1 in 2000 w/v. What are the percentage strength and the amount concentration expressed as mg/mL?

% = ? and mg/mL = ?

2.6 Parts per Million (ppm)

Textbook Problems

Q37 Fluoride supplements should not be taken if the amount of fluoride in the water supply exceeds 0.7 parts per million w/v. Express this ratio in mg/L.

mg/L = ?

Q38 If the concentration of fluoride is 0.25 ppm w/v, how many litres would contain 1 mg of fluoride?

Litres = ?

2.7 Converting Between Expressions of Concentration

Textbook Problems

Q39 A solution contains 20 mL of ethanol in 500 mL of product. Express the concentration as a ratio strength and as a percentage strength.

Ratio = ? and % = ?

Q40 A solid ingredient mixed with a solid vehicle has a ratio strength of 1 in 40. Find the percentage strength and the amount strength expressed as grams per gram.

% = ? and g/g = ?

Q41 Convert the following ratio strengths into percentage strengths:

(a) 1 in 25

(b) 1 in 20

(c) 1 in 50

(d) 1 in 800

(e) 1 in 500

(f) 1 in 2000

(g) 1 in 300

Q42 A liquid ingredient mixed with another liquid vehicle has a concentration of 5% v/v. Find the ratio strength and the amount strength expressed as mL/mL.

Ratio = ? and mL/mL = ?

Q43 5 g of solid ingredient is added to 45 g of a base. Find the percentage strength, the ratio strength and the amount strength expressed as g/g. (Same as Q48.)

% = ?, ratio = ?, g/g = ?

Q44 What is the concentration of the solutions, expressed as percentage strength and ratio strength, when the following amounts of drug Z are dissolved in enough water to produce 125 mL of solution?

(a) 25 g

(b) 50 g

(c) 60 g

(d) 5 g

(e) 7 g

2.8 Heparin & Insulin (Units)

Lecture Problems

Q45 Heparin is available as 1000, 5000, 10000 and 25000 units/ml (1 ml ampules).

You need to give heparin as continuous infusion 13000 units in 35 ml normal saline over 24 hours. What do you withdraw?

2.9 Practice Calculations (Lecture)

Lecture Problems

Q46 Practice Calculation 1

How many grams of sodium chloride are there in one litre infusion of sodium chloride 0.9%?

Q47 Practice Calculation 2

You need to give infusion of calcium gluconate 3 g. You have 10 ml ampules of calcium gluconate 10%. How much do you need to withdraw?

Q48 Practice Calculation 3

5 g of solid ingredient is added to 45 g of a base. Express as percentage, ratio, and as amount of strength g/g.

---

3 Dilutions

In this lecture

1. Simple Dilutions 2 lecture · 2 textbook
2. Stock Solutions & Serial Dilutions 3 lecture · 9 textbook
3. Concentrated Waters 1 lecture · 1 textbook
4. Multiple Dilutions 1 lecture · 12 textbook
5. Triturations 1 lecture · 1 textbook
6. Powder Calculations 1 lecture · 3 textbook
7. Mixing Concentrations 2 lecture · 7 textbook

3.1 Simple Dilutions

Lecture Problems

Q1 You need to prepare 500 ml of 45% solution. What volume of stock solution do you need which when diluted to 500 ml will give 45% solution?

Volume of stock = ? and volume of water = ?

Q2 *Student activity:* 300 ml of 1 in 25 w/v solution diluted to 800 ml. What is the concentration of the diluted product in % and mg/ml?

a. % w/v = ?

---

b. mg/ml = ?

---

c. Ratio strength = ?

Textbook Problems

Q3 100 mL of a 1 in 50 w/v solution is diluted to 1000 mL. Find the concentration of the diluted product as a percentage strength, a ratio strength and an amount strength expressed as mg/mL.

% , ratio and mg/mL = ?

Q4 A product consists of a solid ingredient in a solid vehicle; 2 g of the solid ingredient is contained in 500 g of product. A further 250 g of vehicle is added to the product. Find the concentration of the diluted mixture as a percentage strength, a ratio strength and an amount strength in mg/g.

% , ratio and mg/g = ?

3.2 Stock Solutions & Serial Dilutions

Lecture Problems

Q5 You need to prepare 1 litre of 20% solution from 80% stock solution.

Stock volume = ? and diluent = ?

Q6 You need to prepare one litre of potassium chloride solution 1 in 10,000. You have a stock solution 10 g/L. How much of the stock solution do you need?

Stock volume = ?

Q7 To what volume must 350 ml of 40% w/v solution be diluted to produce a 12% solution?

Final volume = ?

Textbook Problems

Q8 What volume of a 40% v/v solution needs to be used to produce 500 mL of 5% v/v solution?

Volume = ?

Q9 How many millilitres of 90% alcohol when diluted to 135 mL produces 60% alcohol? (Assume no contraction in volume.)

Volume = ?

Q10 What volume of a 1 in 400 v/v solution is needed to produce 5 L of a 1 in 2000 v/v solution?

Volume = ?

Q11 What volume of a 0.6% w/v solution is required to produce 200 mL of 1 in 10 000 solution?

Volume = ?

Q12 What volume of a 1 in 50 solution is required to produce 300 mL of a 0.3% w/v solution?

Volume = ?

Q13 To what volume must 250 mL of a 25% w/v solution be diluted to produce a 10% solution?

Final volume = ?

Q14 How many millilitres of water must be added to 180 mL of 40% v/v solution in order to produce a 5% v/v solution?

Water = ?

Q15 What volume of 100 mg/5 mL solution is required to be diluted to give 200 mL of 75 mg/5 mL solution?

Volume = ?

Q16 Calculate the volume of 0.2% w/v potassium permanganate solution required to produce 1000 mL of a 100 ppm solution.

Volume = ?

3.3 Concentrated Waters

Lecture Problems

Q17 Concentrated waters (rose water, peppermint water, chloroform water) are used to produce single-strength products. They are intended for dilution in a ratio of 1 in 40.

What does "1 in 40" mean here?

Textbook Problems

Q18 Make volumes of single-strength chloroform water (50 mL, 100 mL, 200 mL, 300 mL, 500 mL) from chloroform water concentrate (1 : 40).

a. Concentrate (mL) for each volume = ?

b. Double-strength chloroform water — concentrate for the same volumes = ?

3.4 Multiple Dilutions

Lecture Problems

Q19 *Student activity:* What weight of ingredient is required to produce 500 ml of a solution, when 3 ml diluted to 40 ml, produces a strength of 0.5% w/v solution?

Weight = ?

Textbook Problems

Q20 What weight of ingredient is required to produce 1000 mL of a solution such that, when 2.5 mL of it is diluted to 50 mL with water, it gives a 0.25% w/v solution?

Weight = ?

Q21 What weight of ingredient is required to produce 250 mL of a solution such that 10 mL of this solution diluted to 150 mL gives a 0.1% w/v solution?

Weight = ?

Q22 What weight of potassium permanganate is required to produce 300 mL of solution such that 5 mL of this solution diluted to 250 mL gives a 0.01% w/v solution?

Weight = ?

Q23 What weight of potassium permanganate is required to produce 500 mL of solution such that 25 mL of this solution diluted to 500 mL gives a 0.025% w/v solution?

Weight = ?

Q24 What weight of zinc sulfate is required to produce 150 mL of solution such that 10 mL of this solution diluted to 200 mL gives a 1% w/v solution?

Weight = ?

Q25 What weight of potassium permanganate is needed to produce 300 mL of solution such that if 1 mL is diluted to 20 mL a 0.1% w/v solution is produced?

Weight = ?

Q26 What weight of malachite green oxalate is required to produce 500 mL of solution such that 25 mL of this solution diluted to 4000 L gives a 1 in 2 000 000 solution?

Weight = ?

Q27 What weight of potassium permanganate is required to produce 200 mL of solution such that 50 mL of this solution diluted to 200 mL will produce a 1 in 400 solution?

Weight = ?

Q28 What weight of copper sulfate is required to produce 150 mL of solution such that 30 mL of this solution diluted to 150 mL produces a 1 in 6000 solution?

Weight = ?

Q29 How many millilitres of 0.5% w/v solution are required to produce 250 mL of 1 in 5000 solution?

Volume = ?

Q30 How many millilitres of a 1 in 80 w/v solution are required to make 500 mL of a 0.02% solution?

Volume = ?

Q31 What weight of malachite green oxalate is required to produce 300 mL of a solution such that 25 mL of this solution diluted to 2000 L gives a 0.5 ppm solution?

Weight = ?

3.5 Triturations

Lecture Problems

Q32 You need to prepare 100 ml of the following formula:

Hyoscine hydrobromide	500 mcg
Water to	5 ml

a. Amount of hyoscine in 100 ml = ?

---

b. Minimum weight on the scale is 100 mg. How do you prepare it?

Textbook Problems

Q33 How would you prepare 100 mL of a preparation to the following formula?

hyoscine hydrobromide (micrograms)	500
chloroform water (mL) to	5

(Same as Q32.)

How do you prepare it?

3.6 Powder Calculations

Lecture Problems

Q34 Powders must weigh a minimum of 120 mg. Prepare seven powders each containing 80 mg of drug x.

Drug, diluent and total weight for 1 and 7 powders = ?

Textbook Problems

Q35 Prepare five powders each containing 100 mg of paracetamol.

Weights = ?

Q36 Prepare five powders each containing 10 mg of hyoscine hydrobromide.

How do you prepare them?

Q37 Prepare six powders each containing 0.5 mg of drug.

How do you prepare them?

3.7 Mixing Concentrations

Lecture Problems

Q38 What is the final strength (% v/v) if 200 ml of 40% v/v solution is added to 300 ml of 70% v/v solution?

Final % v/v = ?

Q39 What proportion of 60% v/v solution and 10% v/v solution would produce a 500 ml of 40% v/v solution mixture?

V1 and V2 = ?

Textbook Problems

Q40 What is the final % v/v of a solution if 200 mL of 40% v/v solution is added to 300 mL of 70% v/v solution? (Same as Q38.)

Final % v/v = ?

Q41 When 70 mL of sodium chloride 0.9% w/v solution is combined with 100 mL of 1.7% w/v sodium chloride solution what is the final strength of the solution?

Final strength = ?

Q42 Calculate the volumes of two stock solutions of 100 mg/5 mL and 25 mg/5 mL that must be mixed to produce 500 mL of 75 mg/5 mL solution.

Volumes = ?

Q43 What volumes of two stock solutions of 30% w/v and 60% w/v are required to make 300 mL of 40% w/v solution?

Volumes = ?

Q44 A suspension used to be provided as two strengths, both 100 mg/5 mL and 25 mg/5 mL solutions, which could be mixed to produce intermediate doses. If we want to produce a product that has a strength of 75 mg/5 mL, what proportion of the two mixtures do we need?

Proportion = ?

Q45 Find the proportions of the same two suspensions given in Q44 that would be required to produce a 50 mg/5 mL suspension.

Proportion = ?

Q46 What proportions of 90% v/v and 50% v/v ethanol mixtures would produce a 70% v/v mixture? Assume no contraction.

Proportion = ?

---

4 Moles

In this lecture

1. Moles & Molecular Mass 3 lecture · 2 textbook
2. Amount of an Ion in a Compound 1 lecture · 8 textbook
3. Drug Salts, Bases & Hydrates 1 lecture · 15 textbook
4. Legal Classification Using Molecular Weights 3 textbook
5. Moles & Millimoles 1 lecture · 5 textbook
6. Millimoles in Infusions & Molar Solutions 4 lecture · 7 textbook

4.1 Moles & Molecular Mass

Lecture Problems

Q1 One mole = the relative atomic mass in grams. One mole of any element contains 6×10^{23} atoms (Avogadro's constant). Atomic masses: Ca = 40, Cl = 35.5, Na = 23, C = 12, H = 1.

a. How many moles are in 24 g of carbon?

b. How many moles (and atoms) are in 1 g of carbon?

Q2 Sodium chloride (NaCl) is made up of sodium (Na) and chloride (Cl): $NaCl \rightarrow Na^{+} + Cl^{-}$.

Molecular mass of NaCl = ? and what does 1 mole of NaCl give?

Q3 CO_2: C = 12, O = 16.

Molecular mass of CO_2 = ?

Textbook Problems

Q4 Calculate the molecular weights (K = 39, Br = 79.9, Al = 27, Cl = 35.5, Zn = 65.4, S = 32, O = 16, H = 1, C = 12):

a. Potassium bromide, KBr

b. Aluminium chloride, $AlCl_3$

c. Zinc sulfate monohydrate, $ZnSO_4.H_2O$

d. Aspirin, $C_9H_8O_4$

Q5 For the following drugs calculate the molecular weight and the weight of salt containing 1 millimole of the metal ion:

a. (a) potassium chloride, KCl

b. (b) sodium chloride, NaCl

c. (c) calcium chloride, $CaCl_2.2H_2O$

d. (d) calcium lactate, $C_6H_{10}CaO_6.5H_2O$

e. (e) potassium bicarbonate, $KHCO_3$

f. (f) magnesium chloride, $MgCl_2.6H_2O$

g. (g) sodium citrate, $C_6H_5Na_3O_7.2H_2O$

h. (h) ammonium chloride, NH_4Cl

4.2 Amount of an Ion in a Compound

Lecture Problems

Q6 How many mg of sodium ions are contained in 850 mg tablet of sodium chloride?

Na^+ (mg) = ?

Textbook Problems

Q7 How many milligrams of sodium ions are contained in a 600-mg tablet of sodium chloride?

Na^+ (mg) = ?

Q8 Calcium acetate, $C_4H_6CaO_4$.

a. Molecular weight = ?

---

b. Calcium provided by 300 mg of calcium acetate = ?

---

c. mg of calcium acetate in 1 millimole = ?

Q9 Sodium Fluoride BP drops contain 275 micrograms of sodium fluoride per drop. Calculate the quantity of fluoride ion provided by five drops (Na = 23, F = 19).

Fluoride = ?

Q10 What is the percentage of lithium contained in a 400-mg tablet of lithium carbonate?

% Li = ?

Q11 Lithium citrate, $C_6H_5Li_3O_7.H_2O$ (Li = 7).

a. Molecular weight = ?

---

b. % of Li^+ = ?

Q12 The empirical formula of ferrous fumarate is $C_4H_2FeO_4$.

a. Molecular weight = ?

b. Weight of ferrous fumarate that gives 100 mg of iron = ?

Q13 A syrup contains ferrous fumarate 140 mg/5 mL. How much iron will a patient receive per day if the daily dosage regimen is 10 mL twice a day?

Iron per day = ?

Q14 Calcium gluconate (molecular weight 448.4) tablets each contain 53.4 mg of calcium (Ca = 40.08).

a. Calcium gluconate in each tablet = ?

b. Millimoles of calcium per tablet = ?

4.3 Drug Salts, Bases & Hydrates

Lecture Problems

Q15 What weight of ferrous gluconate contains the same quantity of iron as tablet containing 300 mg of ferrous sulphate (dried)? ($FeSO_4$ = 152, ferrous gluconate = 482, Fe = 56)

Ferrous gluconate = ?

Textbook Problems

Q16 What weight of ferrous gluconate contains the same quantity of iron as a tablet containing 200 mg of ferrous sulfate (dried)?

Ferrous gluconate = ?

Q17 Erythromycin 734, erythromycin ethyl succinate 862, erythromycin stearate 1018. Calculate the amounts of erythromycin ethyl succinate and erythromycin stearate that are equivalent to 250 mg erythromycin.

Ethyl succinate and stearate = ?

Q18 Adrenaline injection contains 100 micrograms of adrenaline (as the acid tartrate) per mL. MW adrenaline 183, adrenaline acid tartrate 333. Calculate the amount of adrenaline acid tartrate in each mL.

Acid tartrate per mL = ?

Q19 Ranitidine Injection BP contains ranitidine 25 mg/mL (as the hydrochloride). Calculate the amount of ranitidine hydrochloride needed to prepare 100 mL. (MW: ranitidine hydrochloride 350.9, ranitidine 314.4)

Ranitidine HCl = ?

Q20 A patient requires 300 mg of quinine (MW 324.4). What weight of quinine hydrobromide (MW 423.3) will provide 300 mg of quinine?

Quinine HBr = ?

Q21 How many grams of salbutamol sulfate will contain the same amount of salbutamol base as 5 g of salbutamol? MW salbutamol 239, salbutamol sulfate $(C_{13}H_{21}NO_3)_2.H_2SO_4$ 577.

Salbutamol sulfate = ?

Q22 BNF: quinine (anhydrous base) 100 mg ≡ quinine bisulfate 169 mg ≡ quinine dihydrochloride 122 mg. Quinine base MW = 324. Estimate the molecular weights of the two salts.

MW of bisulfate and dihydrochloride = ?

Q23 Show that a 300-mg tablet of quinine bisulfate contains less quinine than a 300-mg tablet of quinine sulfate (BNF equivalent for quinine sulfate = 121).

Quinine base in each = ?

Q24 1 L of Paediatric Ferrous Sulfate Oral Solution BP contains 12 g of ferrous sulfate. You need 150 mL, but only ferrous sulfate anhydrous is available. MW ferrous sulfate 278, anhydrous 152. How much anhydrous should be used?

Anhydrous ferrous sulfate = ?

Q25 How much codeine phosphate sesquihydrate (MW 424) is equivalent to 30 mg of codeine phosphate (hemihydrate, MW 406)?

Sesquihydrate = ?

Q26 Glucose intravenous infusion contains 5% glucose expressed as anhydrous glucose.

a. Anhydrous glucose (MW 180) for 500 mL = ?

---

b. Glucose monohydrate (MW 192) for 750 mL = ?

Q27 A syrup contains chloroquine sulfate (MW 436), 68 mg/5-mL dose. Calculate the amount of chloroquine base (MW 319.9) in each 5-mL dose.

Base = ?

Q28 How much chloroquine base (MW 319.9) is equivalent to a tablet containing 250 mg chloroquine phosphate (MW 515.9)?

Base = ?

Q29 How much morphine (MW 285.3) is contained in 50 mL of a 2% solution of morphine tartrate (MW 774.8)?

Morphine = ?

Q30 How much hyoscine (MW 303.4) is contained in a 20-mg tablet of hyoscine butylbromide (MW 440.4)?

Hyoscine = ?

4.4 Legal Classification Using Molecular Weights

Textbook Problems

Q31 Ephedrine HCl can be sold over the counter if the maximum dose ≡ 30 mg ephedrine and the maximum daily dose ≡ 60 mg ephedrine. MW ephedrine HCl 202, ephedrine 165. What are these limits as ephedrine hydrochloride?

Max dose and max daily dose = ?

Q32 Morphine salts are CD Inv POM up to 0.2% (as anhydrous morphine base). MW base 285, morphine HCl 376. Maximum morphine hydrochloride in 200 mL?

Morphine HCl = ?

Q33 Codeine is CD Inv POM up to 2.5% (as base) or 100 mg base per dosage unit. MW codeine base 317, codeine phosphate 406.

a. Max codeine phosphate in 100 mL syrup = ?

---

b. Max codeine phosphate per dosage unit = ?

4.5 Moles & Millimoles

Lecture Problems

Q34 In practice, moles are too big for everyday use, so millimoles are used.

1 millimole = ? mole

Textbook Problems

Q35 How many millimoles each of Na^+ and HCO_3^- are there in a 500-mg sodium bicarbonate capsule? ($NaHCO_3$ = 84)

mmol = ?

Q36 Calculate the number of moles of chloride ion in 500 mg of calcium chloride ($CaCl_2$ = 110).

Chloride = ?

Q37 Calcium phosphate, $Ca_3(PO_4)_2$.

a. Molecular weight = ?

---

b. A powder contains 3.3 g. Millimoles of calcium ion = ?

Q38 Calculate the number of moles of water in 500 mg of $CaCl_2.6H_2O$ (MW 219).

Water = ?

Q39 "Low sodium" means less than 1 millimole of sodium per tablet or 10-mL dose. What is the maximum amount of sodium chloride in a 10-mL dose?

NaCl = ?

4.6 Millimoles in Infusions & Molar Solutions

Lecture Problems

Q40 How many millimoles of sodium are there in 500 ml infusion containing 1.8 mg/ml sodium chloride?

mmol Na = ?

Q41 How many millimoles of sodium are there in a 1 litre infusion containing 1800 mg/litre sodium chloride?

mmol = ?

Q42 How many millimoles of sodium are in one litre sodium chloride infusion 0.9%?

mmol = ?

Q43 *Student activity:* You need to add 35 mmol of potassium chloride to one litre of infusion. You have ampule containing potassium chloride 2 g/10 ml. Calculate how much do you need to withdraw? (KCl = 39 + 35.5 = 74.5)

Volume = ?

Textbook Problems

Q44 How many millimoles of sodium and chloride ions are contained in 1 L of Sodium Chloride Infusion BP (0.9%)? (Same as Q42.)

mmol = ?

Q45 A syrup contains 7.5% of potassium chloride. How many millimoles of K^+ are contained in 1 mL?

mmol K^+ = ?

Q46 Magnesium sulfate injection is 50% magnesium sulfate ($MgSO_4.7H_2O$). How many millimoles of magnesium ion per mL?

mmol/mL = ?

Q47 How many moles of solute are there in 20 mL of a 1 mol/L solution?

Moles = ?

Q48 How much sodium chloride is required to make 25 mL of a 0.5 molar solution? How many moles of sodium ion will there be?

NaCl and Na^+ = ?

Q49 How much sodium fluoride (MW 42) is required to make 250 mL of a mouthwash containing 0.012 mol/L?

NaF = ?

Q50 How many millimoles/L of chloride ions are in Potassium Chloride (0.3%) and Sodium Chloride (0.9%) Infusion BP? (KCl 74.5, NaCl 58.5)

Chloride = ?

---

5 Doses for Adults

In this lecture

1. Solid Dose Units (Tablets, Capsules...) 1 lecture · 22 textbook
2. Oral Liquid Formulations 1 lecture · 13 textbook
3. Dose Volumes Less Than 5 ml 1 lecture · 4 textbook
4. Injections & Units 6 textbook
5. Doses Based on Body Weight 3 lecture
6. Doses Based on Body Surface Area (BSA) 3 lecture

5.1 Solid Dose Units (Tablets, Capsules...)

Lecture Problems

Q1 A prescription requires two capsules of drug X for 5/7. How many tablet should be supplied?

Number = ?

Textbook Problems

Q2 Calculate the total number of tablets to be dispensed if a prescription requires two tablets of drug X three times a day for 7 days.

Tablets = ?

Q3 A prescription requires two suppositories of drug Y daily for 3/7. How many suppositories should be supplied?

Suppositories = ?

Q4 A patient is prescribed two tablets of drug A four times a day for 3 months. How many tablets?

Tablets = ?

Q5 Charcoal sachets: two sachets three times a day for 1/12. How many sachets?

Sachets = ?

Q6 A prescription requires a patient to receive 20 mg domperidone (in tablet form) every 8 hours for 4 weeks. Domperidone is available only as 10-mg tablets. How many tablets?

Tablets = ?

Q7 A scored tablet contains 100 mg of drug X. Prescription: 50 mg three times a day for 10 days. How many tablets?

Tablets = ?

Q8 Drug Z is only available as a 62.5-mg capsule. Prescription: 125 mg four times a day for 1/12. How many capsules?

Capsules = ?

Q9 Drug A is available as a 500-microgram tablet. A patient requires 2 mg four times a day for 3/7 and 1.5 mg for 5/7.

a. Tablets for a 2-mg dose = ?

---

b. Tablets for a 1.5-mg dose = ?

---

c. Total tablets = ?

Q10 Drug B: 500 mg twice a day for 5 days; available as 250-mg capsules. How many capsules?

Capsules = ?

Q11 A patient requires 8 mg of drug X per day for 1 week followed by 7 mg per day for the next week. Drug X is available in tablets of 1 mg and 4 mg. How many tablets of each strength?

Tablets = ?

Q12 Drug A is available as 2.5-mg and 5-mg tablets. Prescription: 7.5 mg TDS for 5 days, then 5 mg QDS for 3 days, then 5 mg BD for 3 days, then 2.5 mg TDS for 7 days. How many of each?

Tablets = ?

Q13 A customer buys a pack of 100 vitamin capsules. The dosage regimen is one capsule three times a day. How many days will the pack last?

Days = ?

Q14 An inhaler contains 200 metered doses. If the patient inhales two doses three times a day, how long will the inhaler last?

Days = ?

Q15 A residential home buys a 1000 pack of senna tablets. Each of 35 residents takes two tablets each night. How long will it last?

Days = ?

Q16 Analgesic suppositories to be used six times a day for 5 days, then three times a day for 3 days. How many suppositories?

Suppositories = ?

Q17 Drug Y: three capsules twice a day for 3/7, two capsules twice a day for 3/7, one capsule twice a day for 2/52. How many capsules?

Capsules = ?

Q18 A patient needs 8 weeks' antimalarial treatment. Drug A: one tablet per week; drug B: two tablets daily. How many tablets of A and B?

Tablets = ?

Q19 One nicotine patch every 3 days. How many patches for 3 months?

Patches = ?

Q20 A sachet of 0.9% saline is used to bathe an eye. Bathe both eyes twice a day for 1/12. How many sachets?

Sachets = ?

Q21 A patient going on holiday for 3 weeks is supplied 168 tablets. Regimen: two tablets four times a day. Enough?

Enough?

Q22 A pack contains 56 tablets. Regimen: one tablet four times a day for 4 weeks. How many packs?

Packs = ?

Q23 A pack of 84 tablets is designed to last 1 month. If the patient takes one dose twice a day, what is the dose?

Dose = ?

5.2 Oral Liquid Formulations

Lecture Problems

Q24 A prescription of indigestion mixture with dosage regimen of 20 ml four times daily for 5 days.

A. How many 5 ml spoonful of mixture required for each dose?

B. Calculate the number of doses.

C. Calculate the total volume of mixture that should be supplied.

Textbook Problems

Q25 Indigestion mixture: 15 mL four times a day for 5 days. How many 5-mL spoonfuls per dose? Number of doses and total volume?

Spoonfuls, doses, volume = ?

Q26 A 100-mL bottle of antihistamine mixture; dosage two 5-mL spoonfuls four times a day. How many days will it last?

Days = ?

Q27 Drug X: 240 mg four times a day; send 100 mL of 120 mg/5 mL mixture. How many days will it last?

Days = ?

Q28 360 mg of paracetamol four times a day for 3 days, supplied as 120 mg/5 mL suspension. Volume to supply?

Volume = ?

Q29 Drug Z: 62.5 mg three times a day for 5 days; available as 250 mg/5 mL suspension. Volume of each dose and total?

Dose volume and total = ?

Q30 Elixir 150 mg/5 mL. Prescription: 225 mg four times a day for 7 days. Volume to supply?

Volume = ?

Q31 How many 5-mL doses are contained in 120 mL of mixture?

Doses = ?

Q32 How many 15-mL doses are contained in 240 mL of mixture?

Doses = ?

Q33 30 mL of mouthwash twice a day for a 14-day course. Volume to supply?

Volume = ?

Q34 Indigestion mixture: 15 mL four times a day for 2/52. How much to dispense?

Volume = ?

Q35 10 mL of laxative solution at night. Volume used in a month?

Volume = ?

Q36 7.5 mL of a mixture four times a day for 3/52. Volume to supply?

Volume = ?

Q37 A 150-mL bottle of cough elixir. Each of two children gets 5 mL four times a day for 3 days. How many doses left?

Doses left = ?

5.3 Dose Volumes Less Than 5 ml

Lecture Problems

Q38 A prescription states: 2.5 ml three times daily for 4/7. What volume of drug should be dispensed?

Volume = ?

Textbook Problems

Q39 A prescription for a young child reads: 2.5 mL three times a day for 4/7. What volume should be dispensed? (Same as Q38.)

Volume = ?

Q40 A very young child is prescribed 2.0 mL of an analgesic suspension three times a day for 4/7. How many mL to dispense?

Volume = ?

Q41 What volume of paracetamol suspension (250 mg/5 mL) would provide a dose of 200 mg?

Volume = ?

Q42 A pack contains 30 mL of a syrup. How many days will it last if the dose is 0.5 mL three times a day?

Days = ?

5.4 Injections & Units

Textbook Problems

Q43 A patient requires an IM injection of 0.5 mg of adrenaline. Adrenaline injection contains 100 micrograms/mL. How many mL?

Volume = ?

Q44 15 mg of morphine sulfate SC every 4 hours for 16 hours; injection 20 mg/mL. Volume per dose and total morphine?

Volume per dose and total = ?

Q45 Nystatin: 500 000 units every 6 hours. Suspension 100 000 units/mL. Total daily dose and volume for 5 days?

Daily dose and volume = ?

Q46 A solution contains 500 units/mL; pack size 25 mL. 2500 units three times a day for 5 days. How many packs?

Packs = ?

Q47 Vitamin A drops contain 5000 units/mL; 40 drops/mL. How many drops give 750 units?

Drops = ?

Q48 Product A: 8 mg vitamin A (retinyl acetate) + 400 IU ergocalciferol. Product B: 2500 IU vitamin A + 10 micrograms vitamin D. Which contains the most vitamins? (Vit A: 1 IU = 0.344 micrograms; Vit D: 40 IU = 1 microgram)

Which product?

5.5 Doses Based on Body Weight

Lecture Problems

Q49 The usual dose of chlorambucil is 150 mcg/kg once a day. How many milligrams should be administered to a person weighing 154 lb?

Dose (mg) = ?

Q50 Normal gentamicin dose is 3 mg/kg/day in three divided doses. What would be the single (8-hour) dose for a patient weighing 165 lb scheduled to receive only 40% of the usual dose (renal impairment)?

Single dose = ?

Q51 A patient weighing 120 lb was administered 2.1 g of a drug supposed to be dosed at 30 mg/kg. Correct, overdose or underdose?

Verdict?

5.6 Doses Based on Body Surface Area (BSA)

Lecture Problems

Q52 BSA = √(Ht (cm) × Wt (kg) / 3600) (average adult = 1.73 m²). If the dose of a drug is 5 mg/m², what would be the dose for a patient measuring 165 cm and weighing 65 kg?

Dose = ?

Q53 The chemotherapy regimen "CAF" during a 21-day cycle is: cyclophosphamide 500 mg/m², D-1; doxorubicin 50 mg/m², D-1; 5-fluorouracil 500 mg/m², D-1, 8. Calculate the dose of each drug per cycle for a patient with a BSA of 1.9 m².

Doses per cycle = ?

Q54 VAD regimen (multiple myeloma), 28-day cycle: vincristine 0.4 mg/d CIVI, D 1–4; doxorubicin 9 mg/m²/d CIVI, D 1–4; dexamethasone 40 mg/d PO, D 1–4, 9–12, 17–20. Total of each drug per cycle for BSA 1.65 m²?

Totals = ?

---

6 Doses for Children

In this lecture

1. Rules Based on Age & Weight 2 lecture · 1 textbook
2. Doses Based on Body Weight 3 lecture · 9 textbook
3. Doses Based on Body Surface Area (BSA) 3 lecture · 5 textbook
4. Other Dispensing Calculations 4 lecture

6.1 Rules Based on Age & Weight

Lecture Problems

Q1 Drug dosage based on age — write the rule:

a. Young's rule

b. Cowling's rule

c. Fried's rule for infants

Q2 Drug dosage based on weight — write Clark's rule.

Clark's rule

Textbook Problems

Q3 BNF: "for most drugs the adult maximum dose should not be exceeded". A drug has a dose of 8 mg/kg and a maximum dose of 300 mg.

Dose for a 10-kg child and a 40-kg child = ?

6.2 Doses Based on Body Weight

Lecture Problems

Q4 Rx Phenytoin 30 mg/5 ml Suspension, 4 mg/kg/day for a 25-lb child.

Daily dose (mg) and volume (ml) = ?

Q5 The usual dose of sulfisoxazole for infants over 2 months and children is 60 to 75 mg/kg. What would be the usual range for a child weighing 44 lb?

Range = ?

Q6 A pediatric patient receives enalaprilat IV every 12 hours: 5 mcg/kg → 55 mcg per dose. Convert to oral enalapril 100 mcg/kg as a single daily dose. A 2.5-mg tablet is crushed and made up to 12.5 mL with sterile water. Dose in mL?

Volume = ?

Textbook Problems

Q7 Pethidine HCl injection 100 mg/2 mL. Child IM dose 0.5–2 mg/kg. Both extremes for a 12-year-old and the volumes.

Range and volumes = ?

Q8 Fluconazole for mucosal candidiasis in children is 3 mg/kg daily. Calculate the dose for a child aged 3 years and a child aged 11 years and suggest a formulation (capsules 50, 150, 200 mg; suspension 50 mg/5 mL, 200 mg/5 mL).

a. 3-year-old

---

b. 11-year-old

Q9 Griseofulvin oral dose for a child is 10 mg/kg daily; suspension 125 mg/5 mL. Dose and volume for a 5-year-old.

Dose and volume = ?

Q10 Using Q3, calculate the oral dose of griseofulvin and the volume of suspension for a 12-year-old child.

Dose and volume = ?

Q11 Co-trimoxazole IV infusion: 36 mg/kg daily in two divided doses. Dose for a 5-year-old child; sterile solution 96 mg/mL.

Dose and volume = ?

Q12 Amikacin: 15 mg/kg daily in two divided doses. Dose for a 6-month-old child; paediatric injection 50 mg/mL.

Dose and volume = ?

Q13 Drug X: oral dose for a 6-month-old of 8 mg/kg twice a day; suspension 40 mg/5 mL. Dose and volume each time?

Dose and volume = ?

Q14 Morphine sulfate for a 6-month-old child by SC injection: 200 micrograms/kg. Quantity for one dose and volume of 10 mg/mL injection?

Dose and volume = ?

Q15 Hydroxychloroquine sulfate oral dose for a child: 6.5 mg/kg daily. Dose for a 7-year-old?

Dose = ?

6.3 Doses Based on Body Surface Area (BSA)

Lecture Problems

Q16 Drug dosage on body surface area: BSA = √(Ht (cm) × Wt (kg) / 3600) (or use nomogram).

a. Child dose from the adult dose = ?

b. Child dose if the dose per m^2 is given = ?

Q17 The usual pediatric dose of ephedrine sulfate is 25 mg/m^2. Calculate the dose for a child weighing 18 kg and measuring 82 cm in height.

Dose = ?

Q18 If the usual adult dose of a drug is 100 mg, determine the approximate dose for a child with a BSA of 0.83 m^2.

Dose = ?

Textbook Problems

Q19 Oral dose of methotrexate for a 5-year-old child who weighs 18 kg. Oral dose 15 mg/m^2 weekly.

Weekly dose = ?

Q20 Calculate the dose of drug X for a 3-year-old boy if the dose is 120 mg/m^2 each day.

Dose = ?

Q21 The dose of drug X is 75 mg/m^2. Dose for a 5-year-old child of average weight?

Dose = ?

Q22 Using Q7, calculate the dose of drug X for a 10-year-old child of average weight.

Dose = ?

Q23 Didanosine 120 mg/m² every 12 hours. Dose and daily dose for a 7-year-old child who weighs 25 kg.

Dose and daily dose = ?

6.4 Other Dispensing Calculations

Lecture Problems

Q24 Rx Theophylline 80 mg/15 ml, give 1.5 mg as directed.

Volume = ?

Q25 How many drops would be prescribed in each dose of a liquid medicine if 15 mL contained 60 doses? The dispensing dropper calibrates 32 drops/ml.

Drops = ?

Q26 How many milliliters of a liquid medicine would provide a patient with 2 table spoonfuls twice a day for 8 days?

Volume = ?

Q27 It takes approximately 2 g of ointment to cover a child's leg. Ointment for total leg eczema, applied twice a day for 1 week. Which size: 15 g, 30 g, 60 g, 120 g?

Size = ?

---

7 Creatinine Clearance

In this lecture

1. Cockroft-Gault & Choosing the Weight 2 lecture
2. Ideal Body Weight (IBW) 3 lecture
3. Adjusted Body Weight (AjBW) 1 lecture
4. Creatinine Clearance & Dosing 2 lecture
5. Adjusting CrCl for Body Surface Area 1 lecture

7.1 Cockroft-Gault & Choosing the Weight

Lecture Problems

Q1 Cockroft-Gault equation — write it and the units.

CrCl (mL/min) = ?

Q2 Which weight is used in the Cockroft-Gault equation?

Weight by BMI = ?

7.2 Ideal Body Weight (IBW)

Lecture Problems

Q3 The ideal body weight (IBW) — write the formulas.

a. Males

b. Females

Q4 What is the IBW of a man who is 174 cm tall and weighs 70 kilograms?

IBW = ?

Q5 What is the IBW for a woman who is 64 inches tall and weighs 70 kilograms?

IBW = ?

7.3 Adjusted Body Weight (AjBW)

Lecture Problems

Q6 AjBW (obese patients, actual weight > IBW) = IBW + 0.4 × (actual body weight − IBW). Calculate the AjBW of a woman who is 157 cm tall and weighs 80 kilograms.

AjBW = ?

7.4 Creatinine Clearance & Dosing

Lecture Problems

Q7 Determine the loading and maintenance doses of gentamicin for a 76-year-old male weighing 190 lb with a height of 6 feet and serum creatinine of 2.4 mg/dL (212 mcmol/L). Loading dose 1.5 mg/kg of IBW; maintenance 1 mg/kg of IBW every 8 hours after the initial dose.

a. Loading and maintenance doses = ?

b. Extra: his creatinine clearance (Cockroft-Gault) = ?

Q8 What is the creatinine clearance for a 40-year-old male patient with S.Cr of 4 mg/dL (354 µmol/L)? The patient is 5 ft 4 in in height and weighs 103 kg.

CrCl = ?

7.5 Adjusting CrCl for Body Surface Area

Lecture Problems

Q9 Adjusted CrCl = BSA / 1.73 × eCrCl. If a patient weighing 120 lb and measuring 60 in. in height has a calculated creatinine clearance of 40 mL per minute, adjust the CrCl based on body surface area.

Adjusted CrCl = ?